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Professor 204

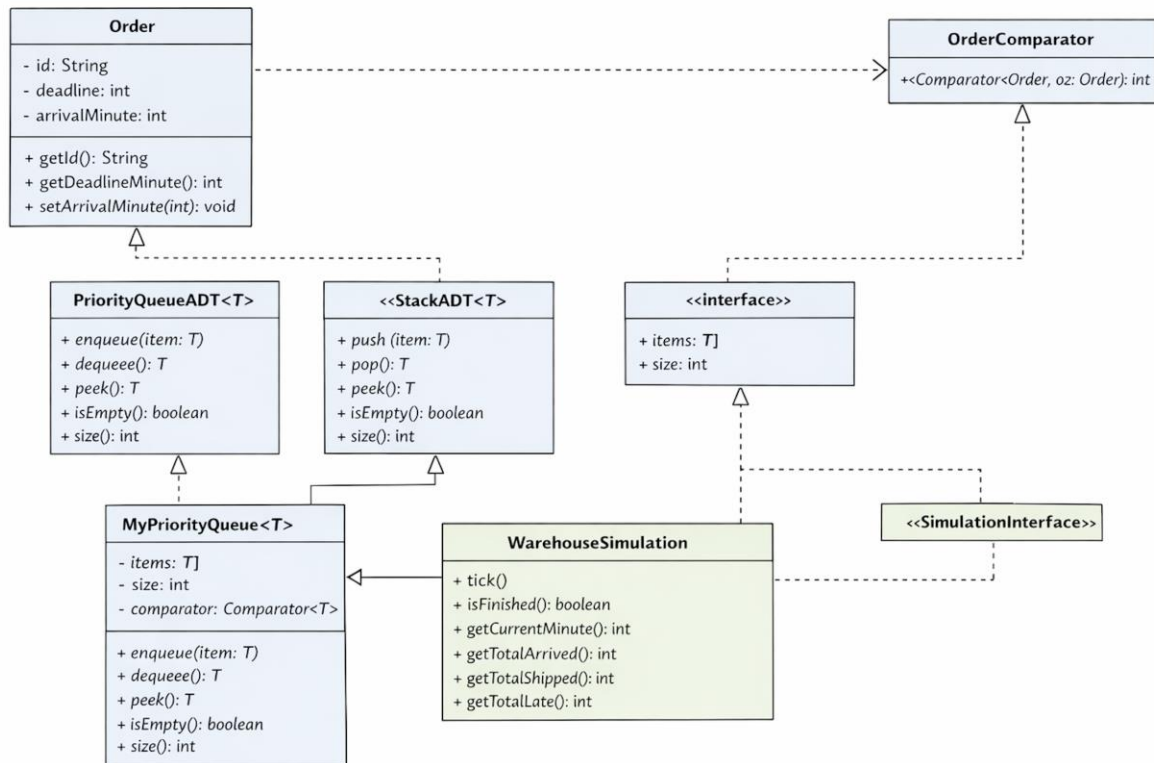
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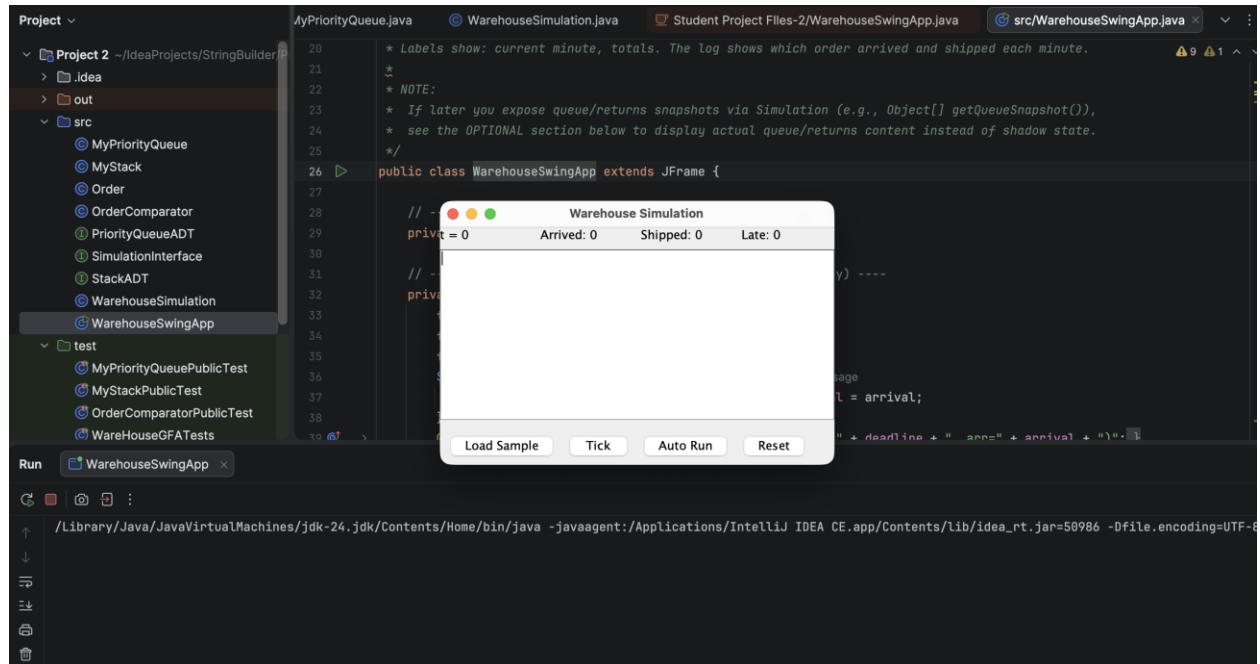
When I began this project, my goal was to implement a warehouse order system that simulates shipments over time. Orders are represented in the Order class, which tracks an order ID, time of arrival, and shipping deadline. The system prioritizes and ships orders based on the earliest deadline, and in the case of ties, it uses arrival time to determine priority. Orders that are shipped after their deadline are considered late and are moved to a return stack for further processing. The core data structures include MyPriorityQueue, a generic priority queue implemented using an array, and MyStack. These structures enforce maximum capacity constraints and support fundamental operations such as enqueue, dequeue, push, and pop. The OrderComparator defines the ordering logic for the priority queue and ensures that the shipping rules are properly enforced.

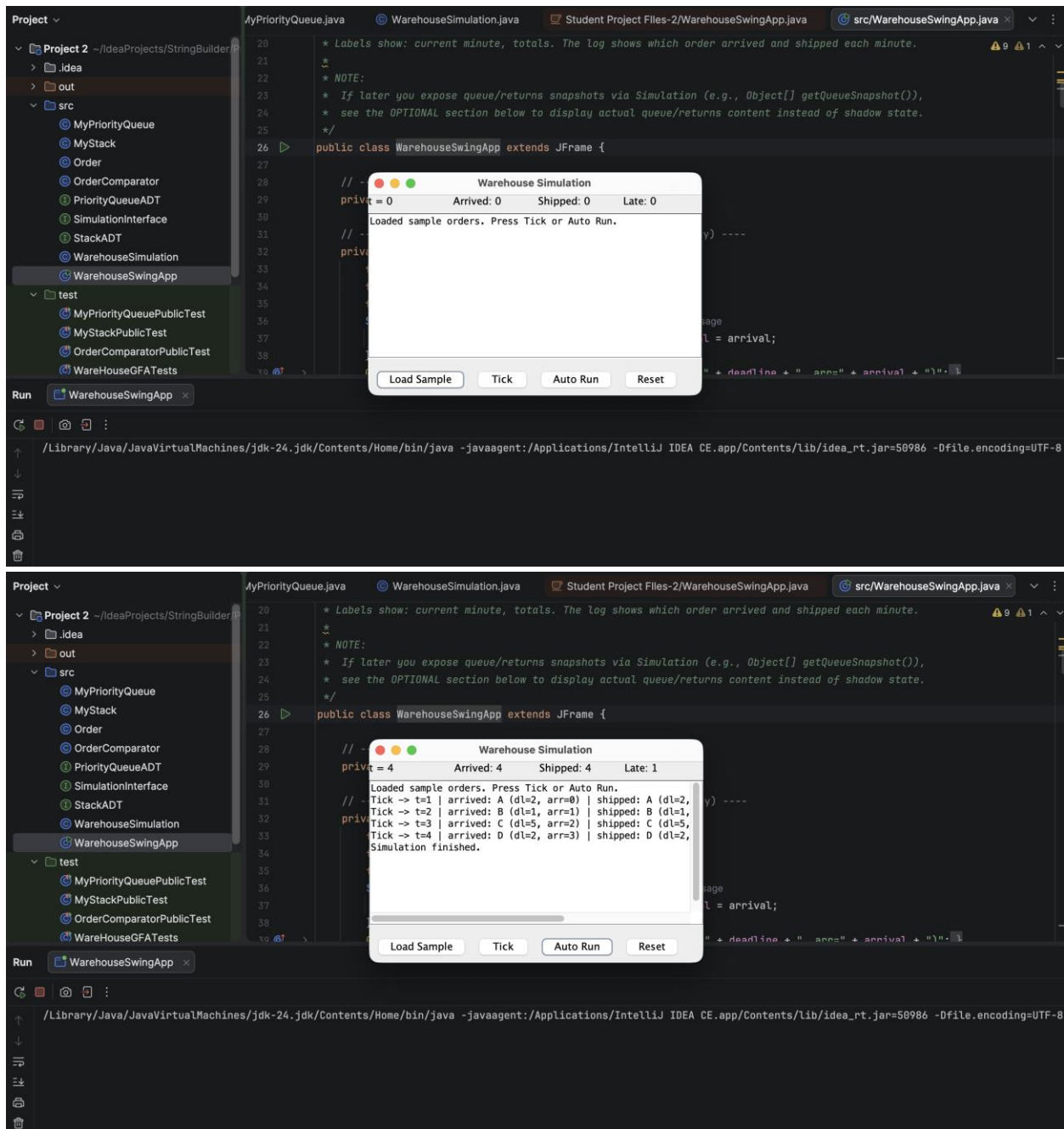
The simulation logic is handled by the WarehouseSimulation class, which implements the SimulationInterface. Each tick of the simulation represents one minute: a new order is released into the queue, the highest-priority order is shipped, and any late shipments are recorded. The system tracks statistics such as the total number of orders released, shipped on time, and shipped late. The provided JUnit tests validate the correctness of the data structures and simulation behavior, ensuring that the system properly handles typical scenarios and edge cases.

UML Diagram



Test Run





Junit Tests

The screenshot displays five test execution panels from an IDE. Each panel shows the test name, its duration, the number of tests passed, and the output of the test.

- MyStackPublicTest**: 8 ms, 4 tests passed (4 tests total). Output: `/Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/bin/java ...`, Process finished with exit code 0.
- OrderComparatorPublicTest**: 7 ms, 2 tests passed (2 tests total). Output: `/Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/bin/java ...`, Process finished with exit code 0.
- WarehouseGFA Tests**: 24 ms, 2 tests passed (2 tests total). Output: `/Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/bin/java ...`, Process finished with exit code 0. A tooltip shows: Overall time: 1 sec 68 ms, Sum time: 24 ms.
- WarehouseSimulationPublicTest**: 17 ms, 2 tests passed (2 tests total). Output: `/Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/bin/java ...`, Process finished with exit code 0.
- MyPriorityQueuePublicTest**: 5 ms, 6 tests passed (6 tests total). Output: `/Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/bin/java ...`, Process finished with exit code 0.